

Architecture of Transformers

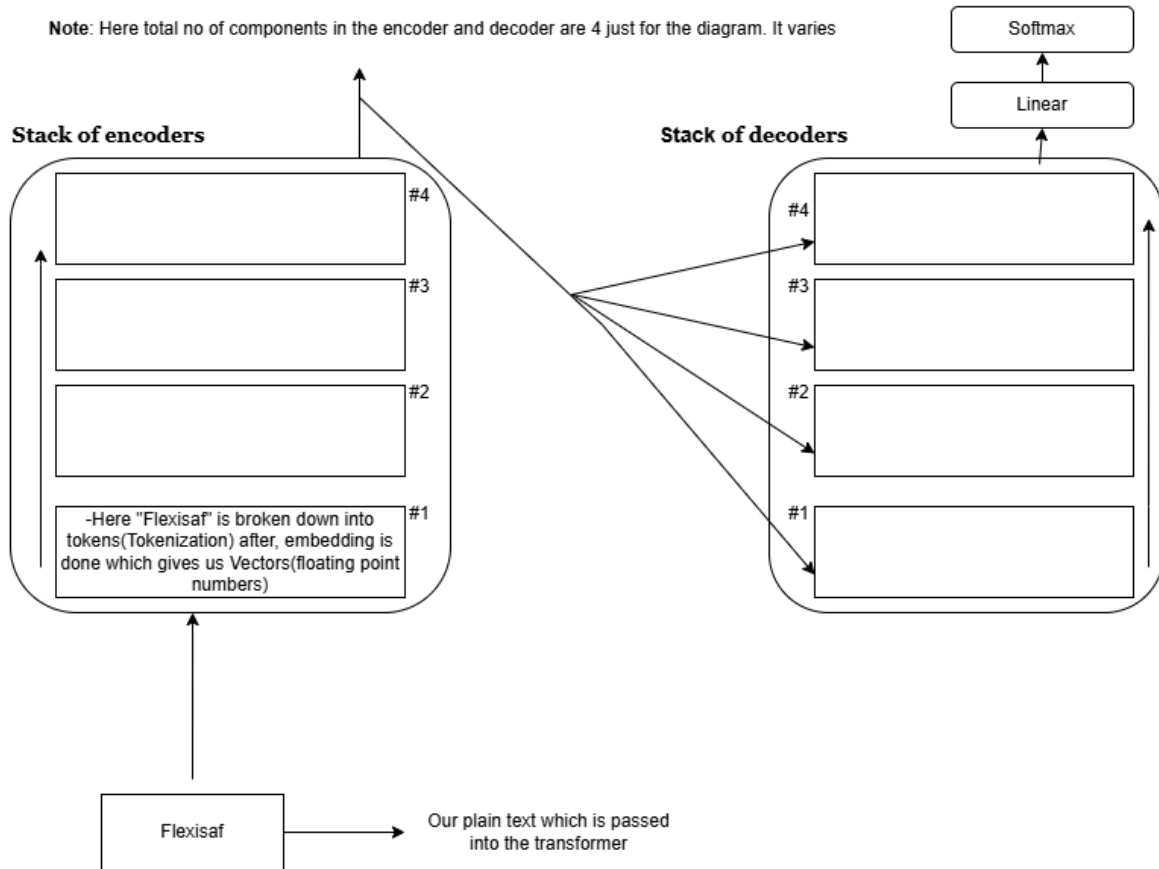
A **Transformer** processes language by first breaking text into tokens, which are the fundamental units of meaning ranging from whole words to fragments of characters. These tokens are converted into numerical vectors that capture their initial semantic value. However, the true magic happens within the transformer blocks, the modular layers that stack to form the model's architecture. Each block contains a mechanism called Self-Attention, which allows the model to look at every other token in a sequence to determine which ones are most relevant to the current word. For example, in the sentence "The bank was closed because of the river," the attention mechanism helps the model link "bank" to "river" rather than "money," dynamically updating the token's meaning based on its context.

Beyond attention, each transformer block includes feed-forward neural networks and layer normalization to refine these contextual signals. These blocks are repeated many times, allowing the model to learn increasingly complex relationships and abstract concepts across long stretches of text. By passing tokens through these successive layers, the Transformer transforms a static list of words into a sophisticated, multidimensional map of intent and information, which is how it ultimately predicts the next most logical token in a sequence with such striking fluency.

Below is a diagram showing us how it works:

System Architecture Of A Transformer

Note: Here total no of components in the encoder and decoder are 4 just for the diagram. It varies



How an Encoder and Decoder look like

